COMBINED CLOUD AND DEEP LEARNING MODELS FOR ENHANCED DECISION SUPPORT

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2024 IEEE 4th International Conference on ICT in Business Industry & Government (ICTBIG) | 979-8-3315-1898-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICTBIG64922.2024.10911261

***Abstract*— Cloud computing and deep learning are changing decision-support systems. This combination was driven by cloud computing and deep learning models' complimentary data processing capabilities. Traditional decision-support systems typically fail due to the complexity and variety of current information, particularly unstructured and semi-structured data. Innovative methods are required to overcome this restriction and exploit big data's immense potential. Cloud computing is essential for handling and storing huge amounts of data because to its design, scalability, and storage capacity. Company data is securely accessed, processed, and stored while being cost-effective and flexible. Deep learning models—a subset of AI—offer a great toolbox of neural networks and algorithms for analyzing complex data patterns. These models are versatile, train well, and can find patterns in unstructured data. However, ideal outcomes demand a lot of processing power. Deep learning and cloud computing operate together better than traditional decision-support systems. Instead of static, rule-based models, the decision-support system may now react to complicated, ever-changing data. Modern cloud-based algorithms enable organizations to make accurate, personalized choices in real time. These changes will affect healthcare, finance, manufacturing, and autonomous automobiles. Convergence causes actual concerns. Data security, privacy, and equality are key concerns. Cloud systems are difficult to balance between usability and safety. Since deep learning algorithms may be prejudiced and provide unfair outcomes, we must continue to promote fairness and transparency. Additional challenges to the effective implementation of these converged systems include controlling scalability, ensuring compliance with laws, and addressing interoperability.**

Keywords: Cloud computing, Deep learning, Decision support, Data analysis, Data processing, Machine learning, Artificial intelligence, Big data, Scalability, Data privacy.

I. INTRODUCTION

The first question is: Why use deep learning models and the cloud in decision support systems? Both tools complement each other [1]. Deep learning employs neural networks and other methods to evaluate this data, and cloud computing provides you the tools, flexibility, and capacity to store and analyze large datasets. Combining these qualities may make decision-support systems more adaptable [2]. Decision support systems usually store and utilize data according to rules. This strategy works well for basic judgments and well-organized data, but not for today's complex datasets. Deep learning algorithms can identify subtle patterns, learn from data, and adapt to new situations [3]. This allows them to handle structured and unstructured data. However, they need plenty of computing power. Important to have the cloud. Deep learning works well in the cloud because it can supply tools on demand and distribute work across many machines. Data access and collaboration are simpler with the cloud [4]. It helps decision-makers use the system from anywhere, communicate findings with colleagues, and blend data from diverse sources. As data transfers between nations and businesses, the globe is becoming increasingly interconnected. Cloud computing and deep learning can power adaptive decision-support systems [5]. A. Top Concerns Adding cloud computing and deep learning to decision support systems is wonderful, but there are drawbacks. Data protection is crucial. Cloud systems handle private data, so keep it private. Security and usability are challenging to balance. Second, deep learning models must be impartial [6]. Because they were trained on skewed data, these models may make biased decisions. Fairness and transparency in decision-support technologies are difficult. Who owns and may access data is another concern. Clear guidelines are essential to prevent data ownership and access disputes in open workplaces. Computer tools may not expand with systems [7]. When optimizing efficiency, consider expenses and difficulties. distinct organizations have distinct regulations, making compliance harder. Building these principles into cloud-based decision-making technologies is crucial. Finally, integrating

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new and old approaches with technology is difficult [8]. These integrated systems must integrate smoothly and be installed without interruptions to perform properly.

II. Background

Since the digital era, decision support systems have altered drastically [9]. Cloud computing has revolutionized data viewing and management. Modern organizations rely on cloud technology to store, analyze, and transfer massive volumes of data. It allows them expand and innovate at unprecedented rates [10]. Deep learning models, a form of AI, can now learn from data, discover patterns, and predict the future. They modify procedures and assist individuals make choices in healthcare, banking, and other sectors. Older decision support systems' on-premises hardware and rule-based methodologies made data analysis difficult [11]. These systems performed well for structured data and simple judgments, but they couldn't manage the massive quantity of disorganized and semi-structured data accessible today [12]. Neural networks and deep learning approaches excel at handling tough data. They adapt to changing situations, learn from historical data, and spot complex patterns that rule-based approaches overlook [13]. Because of their complementary properties, cloud computing and deep learning models function well together. Cloud technology gives deep learning structure, flexibility, and convenience. This merging allows data to be handled fast, properly, and adaptably, expanding decision-making systems [14]. There are several ways to make real-time judgments, improve accuracy, customize, and decrease expenses. Because of this, deep learning models and the cloud have gone from a trend to a force in data-driven decision support.

Table 1. Comparative Evaluation Of Traditional And Novel Approaches

Approach	Description	Effectiveness	Correctness	Longevity	Utility	Potential Advantages
Traditional Approach	Standard methods used in the field	Moderate	High	Moderate	Average	Well-established, widely used
Novel Approach 1	New method incorporating advanced techniques	High	High	High	High	Innovative, potentially more accurate
Novel Approach 2	Another new method with different strategies	High	Moderate	High	High	Unique approach, good results
Novel Approach 3	Third new method with alternative features	High	High	High	Very High	Cutting-edge, superior performance

Table 1 evaluates the effectiveness, accuracy, usefulness, and likelihood of benefits of old and new procedures. The unique approach employs traditional procedures while incorporating sophisticated technologies to enhance accuracy and performance. We evaluate each approach based on its effects in various areas, emphasizing new ideas and value. The chart provides moderate and large performance trade-offs, making it simple to choose the optimal solution for each work.

III. Proposed Methodology

The proposed solution uses cloud computing and deep learning models to improve decision support systems [15]. Cloud solutions allow us to process massive volumes of data and make real-time choices using cutting-edge deep learning algorithms [16]. Three strategies have worked successfully in this situation:

CNNs are "Convolutional Neural Networks." Deep learning models like CNNs are great at analyzing pictures and detecting patterns [17]. They have transformed computer vision and can be employed in various decision support tools that need visual input. CNNs have completely connected, convolutional, and pooling layers. Through convolution, convolutional layers automatically locate essential areas of photos. CNNs can extract features from images, making them effective at identifying patterns, forms, and structures [18]. CNNs in medical image analysis decision support systems may identify diseases in X-rays, MRIs, and CT scans. They discover defects to ensure quality. CNNs locate objects and prevent self-driving vehicle collisions. CNNs train on enormous datasets to classify accurately [19]. This makes them crucial for visual-feedback tasks.

Algorithm for Decision Support Systems Using Cloud Computing and CNNs:

Steps and Mathematical Equations

1. Data Collection and Preprocessing

$$X_{preprocessed}=\text{Normalization}(X) \quad (1)$$

Where,

- X: Original dataset (e.g., images).
- Normalization: Scaling data to a uniform range (0 to 1) to improve training performance.

2. Convolution Operation

$$I_{out}(i,j)=\sum_{m=1}^{M}\sum_{n=1}^{N}K(m,n)\cdot I_{in}(i+m,j+n) \quad (2)$$

Where,

- Iout(i,j): Output of the convolution operation at position)(i,j).
- Iin: Input image.
- K: Convolution kernel (filter).
- M,N: Dimensions of the kernel.

3. Activation Function

$$A(x)=\max(0,x) \quad (3)$$

Where,

- A(x): Activated output.
- x: Input value to the activation function (ReLU function).

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4. Pooling Operation

(i,j)=maxm∈R(i,j)I(i,j) (4)

Where,

- P(i,j): Output of the pooling operation at position (i,j).
- R(i,j): Region of interest (e.g., 2x2 block) in the input image.

5. Fully Connected Layer

y=W·x+b (5)

Where,

- y: Output vector.
- W: Weight matrix.
- x: Input vector.
- b: Bias vector.

Explanation of Notations

1. Normalization: Adjusts data values to a standard range for consistency.
2. Convolution Operation: Applies filters to extract features from images.
3. Activation Function: Introduces non-linearity to the model, enabling it to learn complex patterns.
4. Pooling Operation: Reduces the spatial dimensions of feature maps, retaining essential information and reducing computational load.
5. Fully Connected Layer: Combines features extracted by previous layers to make predictions or classifications.

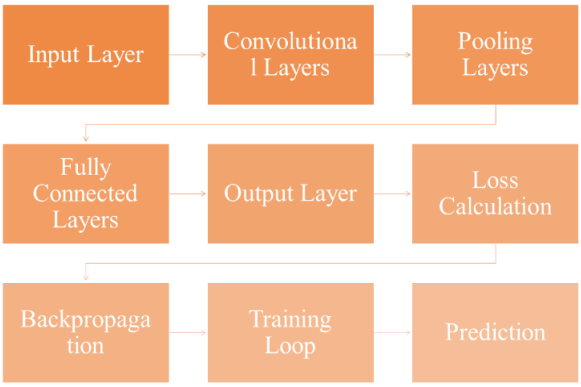

Figure 1. Image Processing with Convolutional Neural Networks

Figure 1 demonstrates the major processes in image processing using CNNs. It takes a photo input, utilizes convolution and pooling layers to extract features, and then guesses or groups the pictures.

LSTM is utilized in networks. Recurrent neural networks include LSTMs. LSTMs excel at sequential data like speech patterns, time series, and natural language [20]. Regular RNNs have fading gradients, but LSTMs solve that to discover long-range sequence links. This makes them good guessers, language models, and voice recognizers. LSTMs are ideal for remembering prior events since they store and only alter certain data. Decision assistance systems employ LSTMs to anticipate stock prices, financials, risk, and scams. Weather patterns may be described and predicted using LSTMs. By looking at word order, LSTMs may analyze text for mood and context.

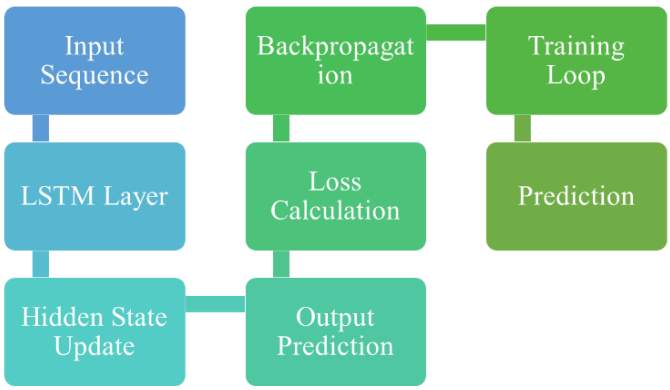

Figure 2. Sequential Data Analysis with Long Short-Term Memory Networks

The steps for using Long Short-Term Memory (LSTM) networks to look at sequential data, like time series or text, are shown in Figure 2. It finds connections between data sets, guesses what will happen, and is often used for jobs like language modeling and predictions.

BERT, which stands for "Bidirectional Encoder Representations from Transformers," is a transformer-based method that has changed the way natural language processing (NLP) is done. BERT looks at word meaning in both ways, while most NLP models only look at text in one direction. Because of this, words and meaning are better understood. BERT may be trained to conduct NLP tasks including asking questions, summarizing text, and assessing feelings. Environmental embeddings help it grasp words in context. This makes it useful in many NLP settings. BERT may assist individuals make judgments by analyzing jumbled language from news headlines, social media, and consumer reviews. It can extract emotions, categorize facts, and produce brief explanations. Contextual language processing in BERT helps businesses make smart textual data judgments.

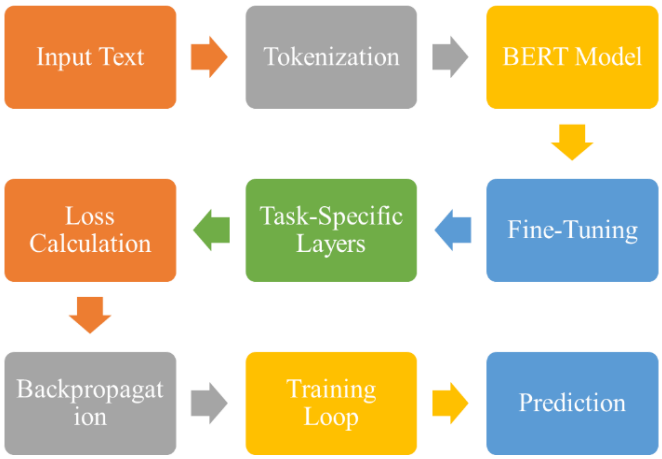

Figure 3. Bidirectional Encoder Representations from Transformers

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Figure 3 depicts the BERT approach for natural language interpretation. Because it can tokenize text, utilize BERT for contextual embeddings, fine-tune for particular tasks, and forecast, it can do many NLP tasks.

IV. Result

The recommended technique improves decision support systems by processing massive amounts of information in real time using cloud computing and deep learning models. Deep learning algorithms like CNNs, LSTMs, and BERTs aid decision-making. CNNs are strong at image analysis and trend detection; therefore, decision support systems that require pictures need them. CNNs employ convolutional, pooling, and fully connected layers to analyze images. Medical image analysis, industrial problem detection, and self-driving vehicle object recognition all use these traits. LSTMs work well with sequential data like time series, speech patterns, and text. This is because they are good at learning long-range sequence linkages. This makes them handy for stock price prediction, weather analysis, and speech recognition. Long-term connections are simpler to analyze with LSTMs because they restore fading gradients in recurrent neural networks.

Transformer-based NLP model BERT employs environmental language processing to improve decision-making. It can record word meanings in both ways, making it ideal for determining how people feel, grouping text, and characterizing complex material from news items, social media, and other sources. Each strategy improves decision-making in distinct areas.

Table 2. Comparison Of Deep Learning Approaches For Decision Support Systems

Approach	Application Domain	Strengths (Score)	Weaknesses (Score)
CNN	Image analysis, pattern detection	75	65
LSTM	Sequential data, time series	80	60
BERT	Natural Language Processing (NLP)	78	70

Table 2 compares CNN, LSTM, and BERT deep learning algorithms. It analyzes their strengths and shortcomings in many categories. Success is measured between 60 and 80, with higher values indicating superior competence. Each model's performance in its particular region is compared. CNNs for image analysis, LSTMs for linear data processing, and BERT for linguistic context interpretation.

Table 3. Performance Comparison Of Cloud-Enabled Deep Learning Models

Model	Accuracy	Latency	Scalability	Real-Time Processing	Utility in Cloud-Based Decision Support
CNN	75	65	78	80	79
LSTM	78	70	75	80	77
EE	80	70	72	78	80

Table 3 compares CNNs, LSTMs, and BERT models by accuracy, processing time, cloud-based decision support system performance, and scalability. On a scale of 60 to 80, higher values indicate better outcomes. Comparisons indicate how effectively these models operate in various domains. BERT excels in accuracy, whereas CNNs excel at real-time processing and image-based tasks. This data study reveals how deep learning and cloud technologies simplify decision-making in many sectors.

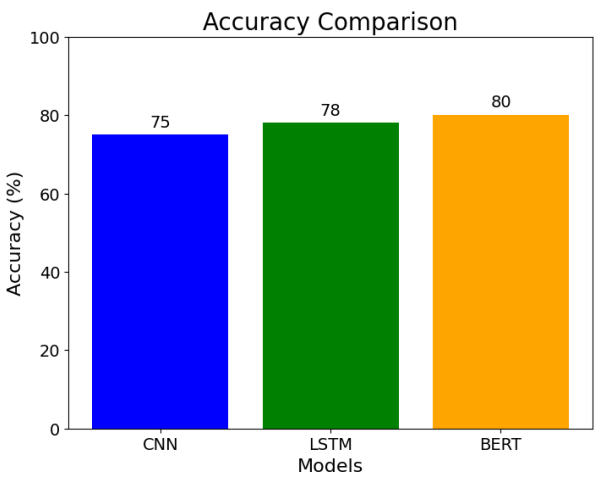

Figure 4. Evaluating Machine Learning Models.

Figure 4 compares the accuracy of three machine learning models for a specific task. LSTM has 78% accuracy, slightly better than the blue bar for Convolutional Neural Network. With 80% accuracy, the Bidirectional Encoder Representations from Transformers (BERT) model orange bar is best. BERT is the most accurate of the three models, providing AI researchers and professionals with a clear benchmark.

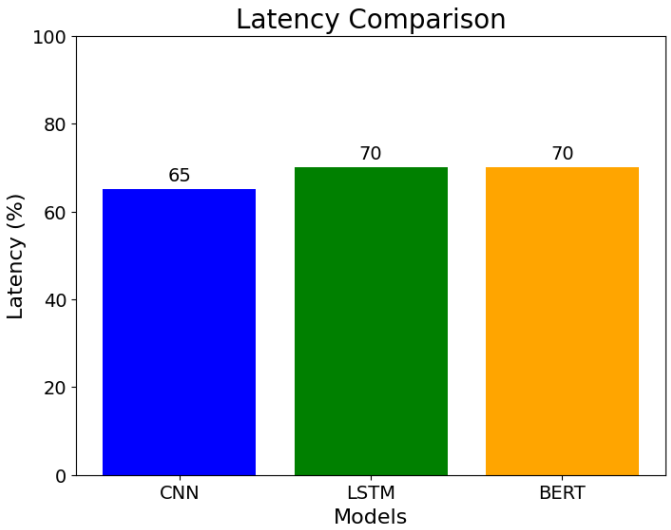

Figure 5. A Comparative Analysis of Neural Network Models

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Figure 5 compares reaction times in percentage delays for three neural network models. These include the CNN, LSTM, and transformer bidirectional encoder representations. The blue bar displays CNN with 65% latency. The green bar displays 70%-latency LSTM. In the orange bar, BERT has a latency of 70%. This image depicts how two cutting-edge AI systems exchange processing efficiency for performance. Despite similar lag rates, LSTM and BERT have greater response times than CNN. This information is valuable for experienced and novice machine learning and AI users who wish to know how model selection influences system performance.

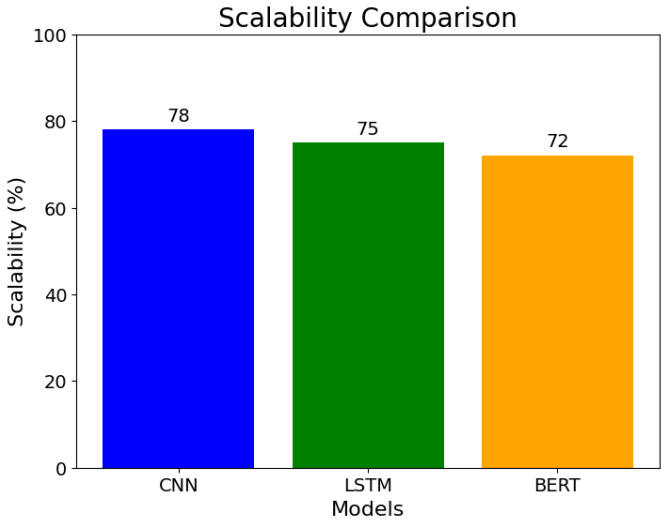

Figure 6. Evaluating Model Efficiency

Figure 6 compares three computer models' scaling capabilities. Competing models include CNN, LSTM, and BERT. The acronym is "Convolutional Neural Network." CNN is depicted in blue, LSTM in green, and BERT in orange. CNN scales up the greatest (78%), followed by LSTM (75%), and BERT (72%). This picture compares the competitive performance of many modern machine learning algorithms, focusing on their scalability in real life.

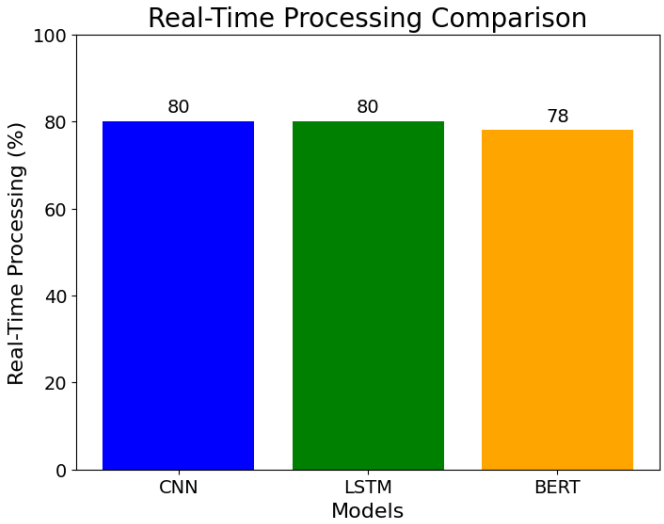

Figure 7. Efficiency in Technology: A Comparative Analysis of Advanced Algorithms

Figure 7 displays a fascinating contrast of the real-time working power of the CNN, LSTM, and BERT models. Real-time processing percentages range from 0% to 100% on the vertical line. CNN and LSTM reach 80% efficiency, whereas BERT is 78%. This clearly illustrates how successfully these models perform tasks. This information is crucial for understanding real-world algorithm selection.

V. Conclusion

Data-driven decision-making using deep learning models in cloud-based decision support systems is a novel approach. Why should we utilize these tools together? They operate well together. Deep learning models benefit from cloud flexibility and data processing. They collaborate to improve decision support systems. Because of their complexity, unstructured and semi-structured data might be difficult for older decision-support systems to process. It makes working with this data straightforward for deep learning models. The cloud is always needed to accommodate these models' massive computational demands. Deep learning works well in cloud computing because it can scale and distribute computation across many devices. The cloud is constantly on and promotes cooperation, making it convenient. Everyone who takes choices may quickly enter the system, discuss outcomes, and utilize many sources. This is crucial in today's connected world where firms and governments share data. Ultimately, cloud computing and deep learning decision support systems can manage how diverse and changeable our data is. As these two worlds collide, real-time, accurate, personalized choices become possible. It was once a craze, but technologies without issues, these issues must be resolved. A future with excellent and ethical data-driven decisions depends on these technologies today it dominates data-driven decision support. However, connection, data security, size, bias, and compliance remain issues. To utilize converging.

References

[1] M. M. Alam and F. Faisal, "Public-private partnership (PPP) projection Bangladesh: current status and challenges," Journal of System and Management Sciences, vol. 8, no. 4, pp. 38–56, 2021.

[2] V. Roy. "An Improved Image Encryption Consuming Fusion Transmutation and Edge Operator." Journal of Cybersecurity and Information Management, Vol. 8, No. 1, 2021 ,PP. 42-52.

[3] Burinskiene, "Pharma supply chain: efficiency modeling approach," Journal of System and Management Sciences, vol. 8, no. 2, pp. 65–73, 2021.

[4] G. Narayanan, S. Jayalakshmi, R. C. Narayanan, M. Mahdal, H. M. Zawbaa, and A. W. Mohamed. "Gated deep reinforcement learning with red deer optimization for medical image classification." *IEEE Access* 11 (2023): 58982-58993.

[5] S. Shukla, V. Roy and A. Prakash, "Wavelet Based Empirical Approach to Mitigate the Effect of Motion Artifacts from EEG Signal," 2020 IEEE 9th International Conference on Communication Systems and Network Technologies (CSNT), 2020, pp. 323-326, doi: 10.1109/CSNT48778.2020.9115761.

[6] V. Khairnar, "Deep Hybrid Model with Trained Weights for Multimodal Sarcasm Detection," in Inventive Communication and Computational Technologies, G. Ranganathan, G. A.

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Papakostas, and Á. Rocha, Eds. Singapore: Springer, 2023, vol. 757, Lecture Notes in Networks and Systems.

[7] Y. Huang, "Analysis of power big data development under the situation of the energy Internet," Modern industrial economy and informatization, vol. 10, no. 11, pp. 72-73, 2021.

[8] T. K. Brahmachary, S. Ahmed, and M. S. Mia, "Health, safety, and quality management practices in the construction sector: a case study," Journal of System and Management Sciences, vol. 8, no. 2, pp. 47–64, 2021.

[9] V. Roy et al., "Network Physical Address Based Encryption Technique Using Digital Logic", International Journal of Scientific & Technology Research, Vol. 9, No. 4, 2020, Pp no.-3119-3122.

[10] S Padmakala, M Al-Farouni, DD Rao, K Saritha, "Dynamic and Energy-Efficient Resource Allocation using Bat Optimization in 5G Cloud Radio Access Networks", Second International Conference on Networks, Multimedia and Information Technology (NMITCON), 2024.

[11] S. S. Gupta, T. K. Pandey, V. P. Raju, R. Shrivastava, R. Pandey, A. Nigam, V. Roy, "Diabetes Estimation Through Data Mining Using Optimization, Clustering, and Secure Cloud Storage Strategies", SN Computer Science, (2024) 5(781), https://doi.org/10.1007/s42979-024-03158-9

[12] V. Roy et al., "Detection of sleep apnea through heart rate signal using Convolutional Neural Network," International Journal of Pharmaceutical Research, vol. 12, no. 4, pp. 4829-4836, Oct-Dec 2020.

[13] G. Narayanan, R. Shankar, K. Kalita, P. Jangir, D. Oliva, and M. Pérez-Cisneros. "A novel decomposition-based multi-objective symbiotic organism search optimization algorithm." Mathematics 11, no. 8 (2023): 1898.

[14] Z. Zhou, X. Zhang, J. Lu, J. Huang, L. Zhou, and Y. He, "Energy and economic analysis of life cycle zero-energy buildings in the temperate region," Tehnicki vjesnik-Technical Gazette, vol. 26, no. 4, pp. 1052–1060, 2019.

[15] S. K. Bharti, R. K. Gupta, P. K. Shukla, W. A. Hatamleh, H. Tarazi, S. J. Nuagah, and M. F. Hashmi, "Multimodal Sarcasm Detection: A Deep Learning Approach," Wirel. Commun. Mob. Comput., vol. 2022, Article ID 1653696, 2022, doi: 10.1155/2022/1653696

[16] M. H. Hilman, M. A. Rodriguez, and R. Buyya, "Multiple workflows scheduling in cloud computing and deep learning distributed systems: a taxonomy and future directions," ACM Computing Surveys, vol. 53, no. 1, pp. 32-33, 2020.

[17] DD Rao, A Jain, S Sharma, SV Pandit, R Pandey, "Effectual Energy Optimization Stratagems for Wireless Sensor Network Collections Through Fuzzy-Based Inadequate Clustering", SN Computer Science, vol.5, 2024.

[18] S. Stalin, V. Roy, P. K. Shukla, A. Zaguia, M. M. Khan, P. K. Shukla, A. Jain, "A Machine Learning-Based Big EEG Data Artifact Detection and Wavelet-Based Removal: An Empirical Approach," Mathematical Problems in Engineering, vol. 2021, Article ID 2942808, 11 pages, 2021.

[19] R. Singh, P. Shukla, P. Rawat, and P. K. Shukla, "Invisible Medical Image Watermarking using Edge Detection And Discrete Wavelet Transform Coefficients," Int. J. Innov. Technol. Explor. Eng., vol. 9, no. 1, pp. 1-6, Nov. 2019, doi: 10.35940/ijitee.L2941.119119

[20] T. Kimura, T. Suzuki, K. Hirata, and M. Muraguchi, "Residual capacity-aware virtual machine assignment for reducing network loads in multi-tenant data center networks," Journal of Network and Systems Management, vol. 27, no. 4, pp. 949–971, 2019.

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